

A tale of two populations: thermal comfort in air-conditioned and naturally ventilated offices in Thailand

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Abstract

A field study of thermal comfort was conducted in Bangkok, Thailand, in which over 1100 office workers responded to a questionnaire while simultaneous physical measurements were taken. In this study we explore whether there is justification for adopting a comfort standard that differs from those developed for office workers accustomed to more temperate climates. Both air-conditioned and naturally ventilated offices were surveyed. Participants cast votes on standard subjective thermal rating scales and these were correlated with temperature indices that variously account for the thermal impacts of humidity, radiant temperature, air velocity, and clothing levels. Following the criteria used in developing a widely adopted thermal comfort standard, it was found that the upper temperature bound for a Thai comfort standard, instead of being the currently accepted level of 26.1 °C, should be as high as 31 °C for office workers accustomed to naturally ventilated spaces, and as high as 28 °C for those accustomed to air-conditioning. Comparing the responses from the naturally ventilated buildings with both those from the air-conditioned buildings and from studies conducted in the temperate regions provides convincing evidence of acclimatization. These and other findings of this study suggest that interior spaces in Thailand can be cooled to a far lesser degree without sacrificing comfort.

Introduction

As the Third World traverses the path of modernization, it appears almost inevitable that buildings will increasingly be air-conditioned to meet the higher standards of amenity that “modernity” implies. One need only travel to the burgeoning cities of the tropics to witness the transformation from the vernacular, open designs utilizing natural techniques for cooling, to closed-box, conditioned architecture. Accepting for the moment that this particular expression of the development process will continue, and recognizing that furnishing air-conditioning to buildings requires significant private and public investment, ongoing costs, and environmental externalities, it behooves us to consider ways to minimize the financial burden.

At the root of air-conditioning use is the yearning for a higher degree of comfort. Air-conditioning design and operation are based upon guidelines developed through thermal comfort research conducted on subjects from temperate climates. These guidelines are closely followed the world over, from climates temperate to tropical.

In this study we explore the question of whether or not there is justification for a different standard of thermal comfort for people accustomed to a climate that is radically different from the temperate bases of comfort guidelines. We do this by assessing the thermal comfort responses of two populations in hot and humid Bangkok, Thailand: those who work in air-conditioned offices and those who work in non-conditioned or naturally ventilated buildings. These two populations give us a wide range of thermal experience in the Thai office, and provide a basis for comparison that can shed light on energy conservation opportunities.

Methods

In this Section we describe the buildings and how we chose them, followed by our methods for conducting the field survey and carrying out the analysis.

Building selection

The criteria for selecting buildings for the field study were as follows:

- located in Bangkok, the capital city of Thailand, where the majority of commercial buildings are;
- modern buildings not more than ten years old;
- both air-conditioned (AC) and naturally ventilated (NV) buildings;
- regular office desk work the primary activity of the building occupants;
- a range of ages and both sexes represented in office workers.

The two air-conditioned buildings are of modern high-rise design, one a head office for a bank, the other a multiple-client building. The two naturally ventilated buildings are contemporary medium-rise government buildings housing ministerial and departmental offices. All buildings are located within ten kilometers of one another in downtown Bangkok.

Data collection

Thailand experiences three distinct seasons in a year. The studies reported in this paper were carried out in each of two seasons: during the hot season (in April) and the wet season (in July) of 1988. (The third season is the cool season, which was not as relevant to a study of cooling.) Each of the four study buildings was visited in both seasons. Data were typically collected at each building over one work-week per season.

The questionnaire consisted of a section of subjective ratings on a variety of thermal scales, followed by a section on recent food and beverage consumption, separate clothing lists for men and women,

and concluded with a section on demographic factors. Subjective ratings employed the seven-point ASHRAE Thermal Sensation Scale shown in Fig. 1. Respondents were asked to mark the scale at any one of the seven points or the mid-points in between them (i.e., at any "tick mark"). The respondents were also asked the question, "I would like to be warmer (1), no change (0), cooler(-1)", otherwise known as the three-point McIntyre Thermal Preference Scale. Two further seven-point scales specifically addressing perceptions of airflow and humidity conditions were also used. The questionnaire was translated into the Thai language from English and scrutinized for accuracy and cultural appropriateness by Thai social scientists with facility in both languages and experience in human subject surveys*.

The measured physical quantities were dry-bulb temperature, relative humidity, globe temperature, and air velocity. The globe thermometer was fashioned from a thermister and a 38-mm-diameter ping-pong ball painted flat grey. The dry-bulb thermister was shielded by a cylinder of reflective foil. Air velocity was measured with a hot-film anemometer. All readings were gathered using a datalogger that

*While the translations were, by all accounts, accurately rendered, there was no rigorous validation of the rating scales in Thai nor is the author aware of any previous attempts to cross-culturally verify them. The use of the translated scales in this analysis is predicated on a common interpretation across cultures.

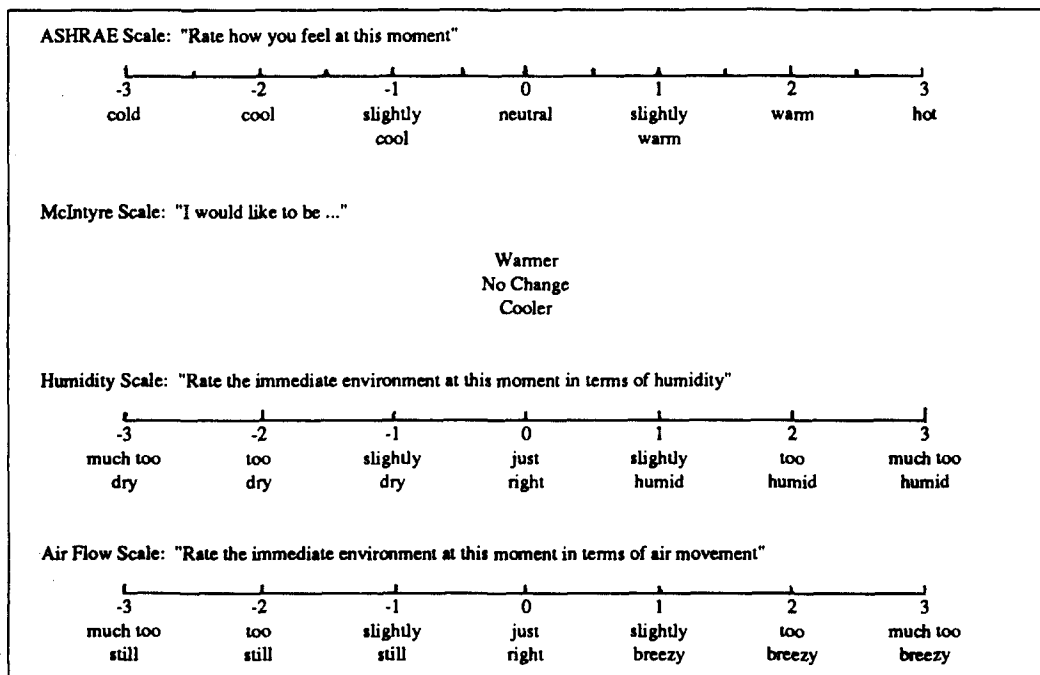


Fig. 1. Subjective rating scales.

stored ten-second readings on magnetic tape. The datalogger, tape recorder, and battery (for the hot-film anemometer) were all contained within, and the temperature and humidity sensors were attached to, a wooden box with a handle similar in size and shape to a standard tool box (see Fig. 2). The hot-film anemometer was detached from the wooden box but connected by a two-meter cord. As is evident from Fig. 2, the sensors were attached vertically to maximize exposure to room air and far enough apart to minimize interference with each other. Data for outdoor weather conditions were gathered from measurements made in the city center by the Royal Thai Meteorological Department.

Conduct of the survey

Teams of two or three typically carried out the survey, with one member taking the physical measurements and one or two handing out and collecting the questionnaires. The latter would approach prospective respondents and ask if they had been seated at that spot for at least 15 minutes; those who replied affirmatively received the questionnaire, the others did not. The questionnaire came with a cover letter explaining the project and the auspices under which it was being carried out, along with general directions for answering the questions. Confidentiality was confirmed and disclosure of respondent's name was optional. Participation by any individual was solicited only once per season, but there was no corresponding effort to exclude people from participating in both seasons. Survey teams sought the participation from a roughly equal proportion of men and women in a range of age and job positions and, to the extent possible, those from different zones and floors of each building.

Measurements of the thermal environment were taken at each workstation following, or in some cases during, the completion of the questionnaire, but usually within five minutes of one another. The wooden box was placed on or very near the desk

where the respondent was seated for at least one minute prior to starting a data sweep. A unique code number for each response was entered into the datalogger and also written on the questionnaire, along with the starting time of the data sweep to assure proper matching of data sets later. The hot-film anemometer wand was held at the subject's torso level, as close to the respondent as decorum allowed (i.e., 0.5 m at a minimum) on the side that intercepted the strongest discernible airflow impinging on the subject. A tell-tale made of thread was used to determine airflow direction. After four minutes of data collection, the wooden box was shifted to the next workstation. Care was taken to allow the equipment to equilibrate when moving to zones with different temperatures.

Assumptions and calculations

Questionnaire data were numerically coded to facilitate statistical analysis. Individual clothing articles indicated in the survey responses were converted into their respective thermal insulation values (I_{comp}) in units of clo (1 clo = 0.155 m²C/W) as tabulated by McIntyre [1]. The overall clo value for each subject's entire clothing ensemble was then determined using the following empirical formulae, also from McIntyre [1],

$$\begin{aligned} I_{\text{clo, men}} &= 0.113 + 0.727 \sum I_{\text{comp}} \\ I_{\text{clo, women}} &= 0.05 + 0.77 \sum I_{\text{comp}} \end{aligned} \quad (1)$$

Metabolic heat production was not directly measured, but since respondents were carefully prescreened to have been seated for at least 15 minutes, their metabolic rate was assumed to be 1.1 met (1 met = 58 W/m²), which is the typical level given for light office activities [2]. Later computation of various comfort indices required determining the body surface area (A_{Du}) of each subject in square meters based on their reported weight (W) and height (H) (in kilograms and meters, respectively) using the Dubois formula [1],

$$A_{\text{Du}} = 0.202 W^{0.425} H^{0.725} \quad (2)$$

Mean radiant temperature (MRT) was calculated as prescribed in the 1984 ASHRAE Systems [3]. A program adapted from the Gagge two-node model [4] was used for calculating two environmental indices that are based on analysis [5] of the thermoregulatory response of the body to thermal stress (described in the next Section).



Fig. 2. Data acquisition system for physical measurements.

Results

Profile of the sample

The total sample of responses numbered 1146 drawn from office workers in four buildings* during each of two seasons. Of these, 669 were women and 476 were men. Six hundred responses were obtained in the hot season and 546 in the wet season. Nobody was surveyed more than once per season, but some portion of the respondents participated in the survey in both seasons[†]. Two-thirds of the sample were from the air-conditioned buildings (770) while the rest (376) were taken from naturally ventilated buildings. The age of the sample ranges from 18 to 75 years and has a mean of 32. Respondents from the AC and NV buildings show a similar age distribution though the former tend towards a slightly younger age.

The distribution of measured physical data by building type is shown in Table 1. Clo values from the AC buildings average 0.56 clo whereas from the NV buildings the average was 0.49 clo. The average Dubois body surface area (not shown in Table 1) for the entire Thai sample was 1.56 m² with a standard deviation of 0.17 m². Air temperatures ranged from a low of 19.5 °C in an air-conditioned building to a high of 34.2 °C in a naturally ventilated building, averaging around 26 °C for the sample with little difference between the hot and wet seasons. Figure 3 shows the distributions of dry bulb temperatures measured in the two buildings types. Vapor pressures averaged 16.9 Torr, again with little seasonality. Air-conditioned buildings had an average air velocity of 0.13 m/s while naturally ventilated buildings experienced higher airflows of 0.33 m/s on average. Because the latter buildings also utilized local fans, air velocities at the workstation were recorded as high as 2.25 m/s.

From these data we calculated the ASHRAE effective temperature (ET*), defined as that temperature at 50% relative humidity, and with mean radiant temperature equal to air temperature, that would produce the same thermal sensation as the actual environment. In other words, ET* normalizes temperatures for humidity and radiation thereby facilitating comparisons of various parameters with a single index. The resultant ET* averaged 27.5 °C for the entire sample. Figure 4 is a relative frequency

TABLE 1. Distribution of physical data

	Air-conditioned	Naturally ventilated
Sample size	770	376
Clothing (clo)		
average	0.56	0.49
standard deviation	0.12	0.10
minimum	0.24	0.24
maximum	1.19	0.72
Air temperature (°C)		
average	23.7	30.8
standard deviation	1.6	1.8
minimum	19.5	23.5
maximum	31.3	34.2
Vapor pressure (Torr)		
average	13.2	24.3
standard deviation	2.7	1.6
minimum	6.9	7.6
maximum	26.1	28.4
Air velocity (m/s)		
average	0.13	0.33
standard deviation	0.04	0.27
minimum	0.09	0.09
maximum	0.88	2.25
ET* (°C)		
average	24.7	33.0
standard deviation	1.8	1.6
minimum	20.5	24.0
maximum	34.0	36.0
SET* (°C)		
average	24.3	31.5
standard deviation	2.0	2.1
minimum	18.1	24.4
maximum	34.3	35.4

distribution of ET* disaggregated by building type. The bi-modal separation of the data between air-conditioned and naturally ventilated buildings is clearly evident.

The standard effective temperature (SET*) is an extension of ET* in that it normalizes for air velocity as well as the two personal variables, clothing insulation and metabolic rate. Standard clothing insulation values are based on metabolic rate. Thus, SET* is defined as the value of an isothermal enclosure with radiant temperature equal to the air temperature, at 50% relative humidity, and air velocity of 0.1 m/s, in which a person with standard clothing for the actual activity level would have the same heat loss at the same mean skin temperature and the same skin wettedness as he or she does in the actual environment with the actual clothing insulation after one hour of exposure. Figure 5 depicts the relative frequency distribution of SET* by building type.

Since respondents in this survey were pre-screened for "standard" activity levels (seated for at least 15 minutes at a desk), SET* differs from

*One additional building served in a single-day pilot study in the hot season and the 25 responses from that building are included in the analysis.

[†]For reasons of confidentiality, participant names were not tracked and therefore an exact figure of multiple-season respondents cannot be calculated.

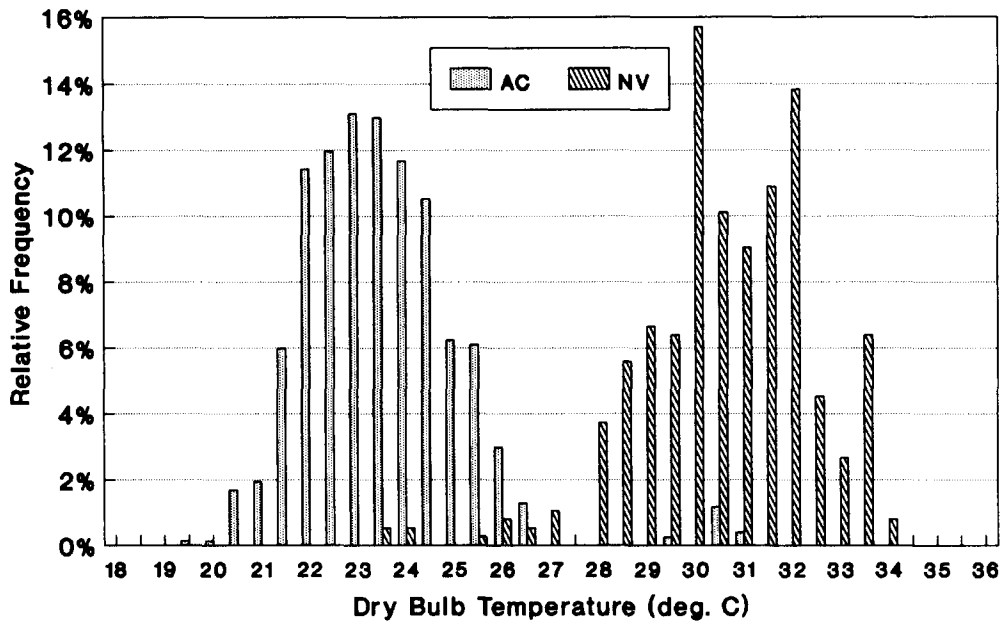


Fig. 3. Temperature.

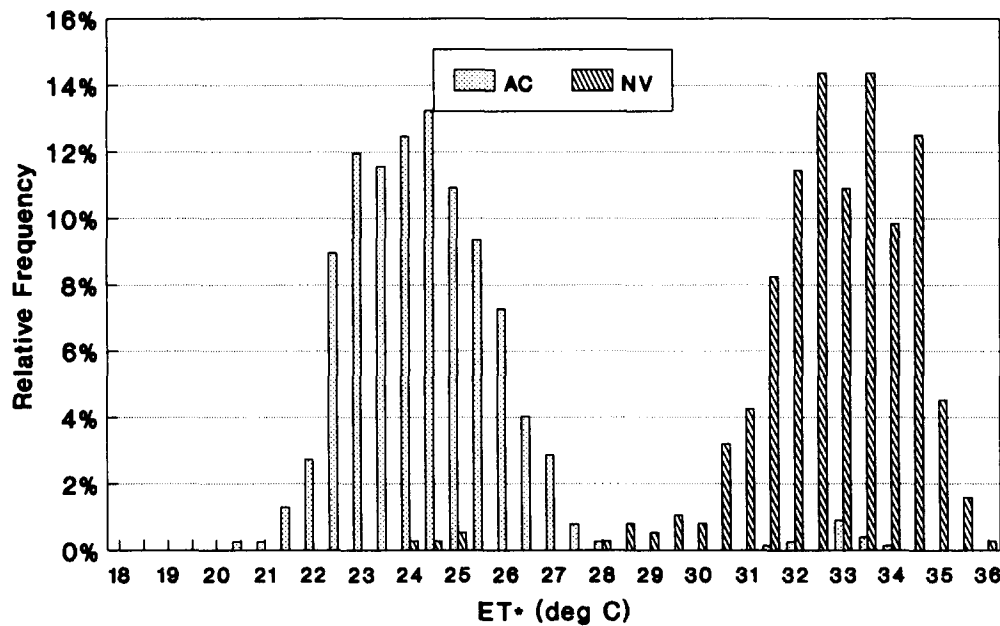


Fig. 4. Effective temperature.

ET* due to air velocities and non-standard clo levels only. Figure 6 depicts the distribution of clo values found in the two building types. Clothing insulation levels were generally lower in the NV buildings by an average 0.07 clo units (see Table 1). This result also confirms our qualitative observation of differing standards for office attire between the building types, in which the NV buildings were occupied solely by government civil servants, and the AC buildings by private sector workers. In Thailand, the business sector has adopted Western formal dress in conforming to international norms, while civil servants

tend to wear more traditional Thai clothing ensembles or dress that would be considered more informal than that of the private sector.

Air speed, the other factor contributing to the difference in the sample's calculated SET* and ET* indices, was likewise different between the two building types. Figure 7 shows a relative frequency plot of air movement at different velocities. While the overwhelming majority (i.e., greater than 90%) of measurements in the AC buildings registered air velocities around 0.1 m/s, in the NV buildings air velocities were much more widely distributed and

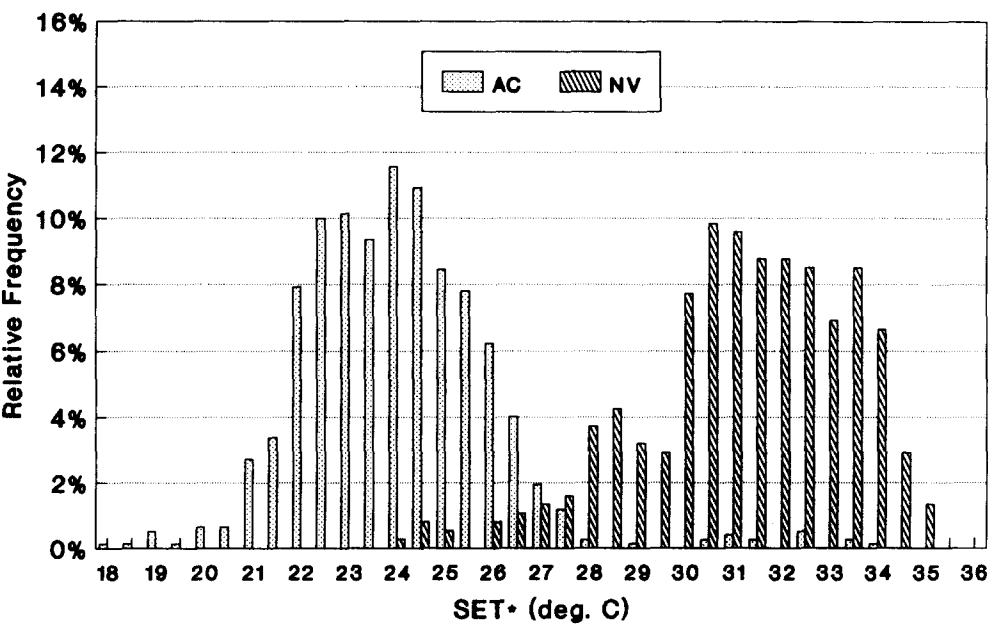


Fig. 5. Standard effective temperature.

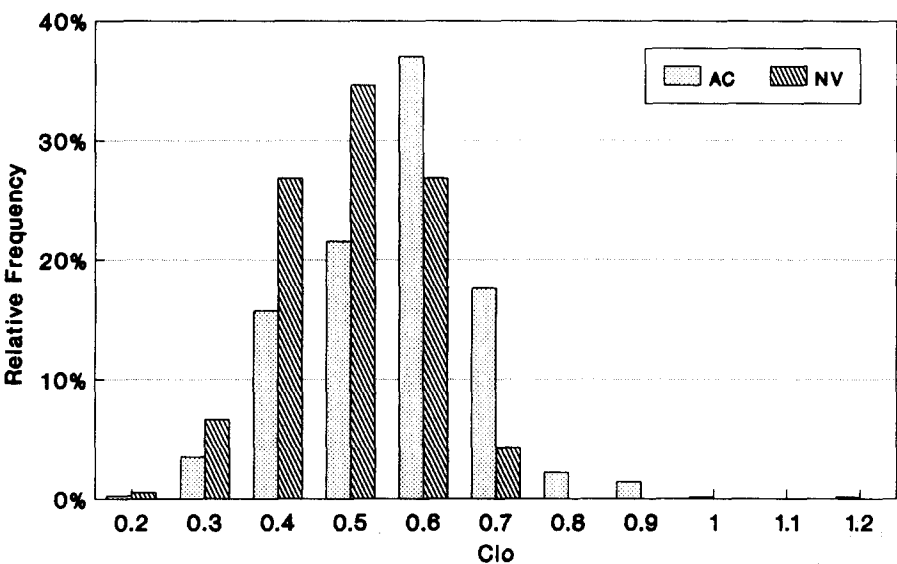


Fig. 6. Clothing insulation value.

generally higher by an average 0.2 m/s. Since it corrects for air speeds that deviate from 0.1 m/s, the SET* index diverges from ET* more for respondents in the NV buildings than those in the AC buildings. Referring to Table 1, the average SET* for the NV building sample is 1.5 °C lower than the average ET*, whereas for the AC building sample, the shift is down only 0.4 °C.

Comparing Figs. 3–5 (shown with identical axes), one can observe the transformation of the distributions as temperature is normalized for other thermal factors. In moving from dry-bulb temperature in Fig. 3 to ET* in Fig. 4, the AC and NV temperature

distributions are shifted roughly 2 °C higher, primarily reflecting the high humidities found in Bangkok. By further incorporating the effect of clothing insulation and air velocities in generating the SET* distributions shown in Fig. 5, a leveling and widening of the distributions is evident. This effect is particularly prominent for the NV distributions. After accounting for the impact of all thermal variables, both environmental and those related to personal choice, the sample displays a remarkable spread of thermal conditions from which the Thai office workers delivered their subjective responses, the subject to which we now turn.

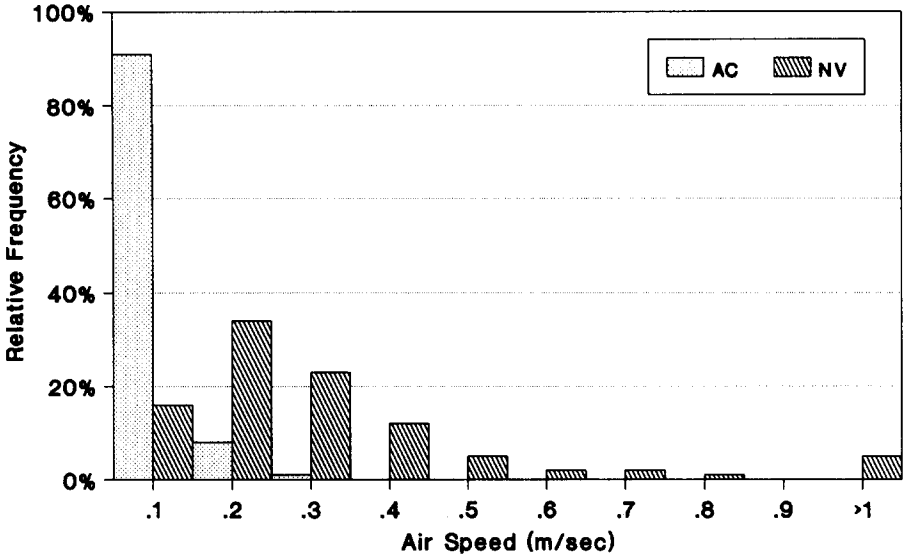


Fig. 7. Air movement.

Distribution of ASHRAE and McIntyre scale responses

The survey participants cast their votes on the seven-point ASHRAE Thermal Sensation and three-point McIntyre scales in response to the immediate conditions at their desks. The distributions of votes disaggregated by building type for both scales are shown in Figs. 8 and 9, respectively. Almost 35% of the votes were cast in the ASHRAE scale zero category (e.g., “neutral”) and three-quarters voted within the central three categories (between “slightly cool” and “slightly warm” or -1 and 1 on the scale). Few people chose to indicate their thermal sensation in the half-steps between whole-numbered categories. The pattern of votes on the ASHRAE

scale from the AC and NV buildings each assume the classic normal distribution, albeit shifted upwards along the scale. Nearly 90% of the respondents in AC buildings selected between “slightly cool” and “slightly warm,” whereas slightly over 50% of the NV building respondents did so.

Responses to the McIntyre scale (graphed in Fig. 9) overall were 42% preferring “no change,” 52% for “cooler,” and 6% for “warmer.” Seventy-eight percent of the NV votes fell into the “cooler” category, whereas the fraction was 40% in the AC case. “No change” was the stated preference of 51% in the AC buildings, but of only 21% in the NV buildings. A surprising 2% voted to be warmer in the NV buildings where temperatures never fell

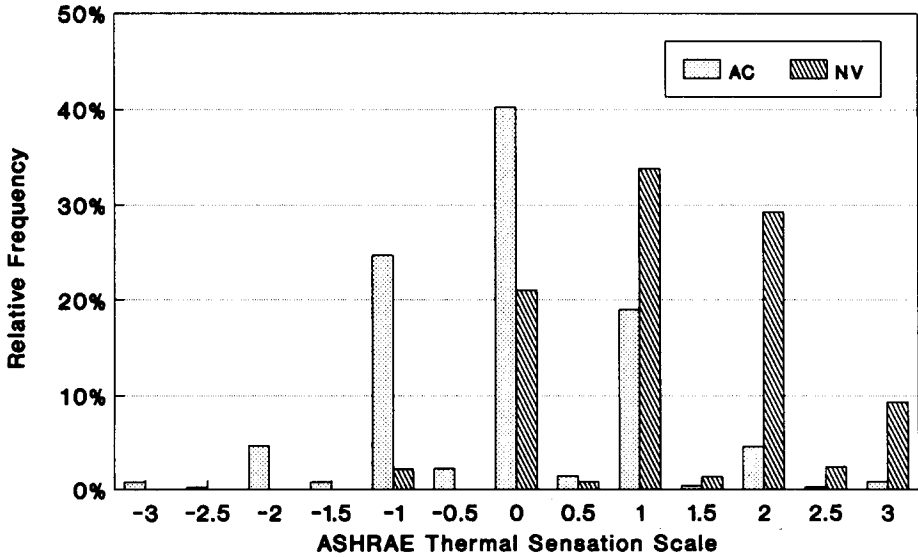


Fig. 8. ASHRAE scale votes.

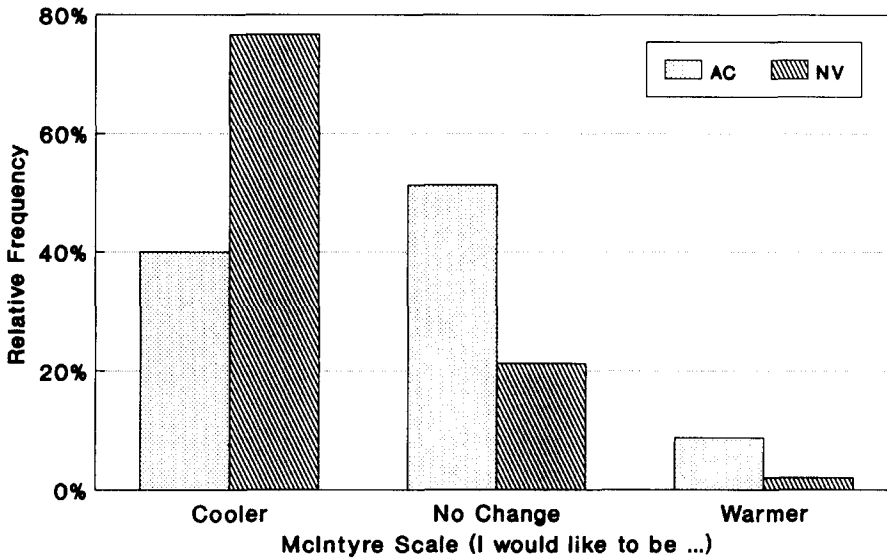


Fig. 9. McIntyre scale votes.

below 24 °C, but misinterpretation of the question by this small fraction of respondents cannot be ruled out.

The scale votes are, of course, taken in response to thermal conditions and therefore are most meaningfully displayed in juxtaposition to a thermal index. In Tables 2 and 3, thermal responses on the ASHRAE scale are cross-tabulated with SET* for the respective building types. These Tables show the percentage of votes at each scale category within 1 °C SET* ranges (i.e., row-wise percentages). Row and column percentage totals are also shown with the number of responses making up each in parentheses. Not unexpectedly, moving from lower to higher SET* levels along each ASHRAE scale category produces

a somewhat normally shaped progression of percentages of votes, each successive category peaking at higher SET*.

The patterns of voting in relation to SET* are shown for the McIntyre scale in Tables 4 and 5 for the AC and NV buildings, respectively. While the general voting trends resemble those on the ASHRAE scale, the individual category patterns follow a more monotonic rising or falling progression.

Regression analysis

Simple linear regression was performed on the ASHRAE scale responses as a function of either ET* or SET* (aggregated into 0.5 °C bins) to determine the strength of the relationship between

TABLE 2. Cross-tabulation of SET* vs. ASHRAE scale in air-conditioned buildings

SET*	% ASHRAE scale thermal sensation votes ^{1,2}							Row totals
	-3	-2	-1	0	1	2	3	
18-20	0	17	17	50	17	0	0	1.5 (12)
21	0	8	39	42	8	4	0	3.4 (26)
22	1	10	39	41	6	1	1	11.3 (87)
23	1	8	30	47	10	3	1	20.1 (155)
24	0	3	25	52	18	2	1	20.9 (161)
25	3	3	27	38	23	4	2	19.4 (149)
26	0	1	18	34	42	6	0	14.0 (108)
27	0	0	7	44	39	11	0	6.0 (46)
28	0	0	9	27	46	18	0	1.4 (11)
29-34	0	7	0	13	7	60	13	1.9 (5)
Column totals	0.8	4.9	25.5	42.5	20.4	4.9	1.0	100
	(6)	(38)	(196)	(327)	(157)	(38)	(8)	(770)

¹Percentages are calculated by row, e.g., within each SET* category.

²Numbers in parentheses are the total number of votes in the respective column or row.

TABLE 3. Cross-tabulation of SET* vs. ASHRAE scale in naturally ventilated buildings

SET*	% ASHRAE scale thermal sensation votes ^{1,2}							Row totals
	-3	-2	-1	0	1	2	3	
24-26	0	0	11	56	22	11	0	2.4 (9)
27	0	0	0	56	33	11	0	2.4 (9)
28	0	0	15	30	30	15	10	5.3 (20)
29	0	0	0	21	50	25	4	7.4 (28)
30	0	0	0	28	38	28	7	10.6 (40)
31	0	0	1	32	34	27	6	19.4 (73)
32	0	0	2	15	35	36	12	17.6 (66)
33	0	0	2	17	33	36	12	15.4 (58)
34	0	0	2	5	32	35	26	15.2 (57)
35	0	0	0	0	31	44	25	4.3 (16)
Column totals	0 (0)	0 (0)	2.1 (8)	21.0 (79)	34.6 (130)	30.6 (115)	11.7 (44)	100.0 (376)

¹Percentages are calculated by row, e.g., within each SET* category.²Numbers in parentheses are the total number of votes in the respective column or row.

TABLE 4. Cross-tabulation of SET* vs. McIntyre scale in air-conditioned buildings

SET*	% McIntyre scale votes ^{1,2,3}			Row totals
	"Cooler"	"No change"	"Warmer"	
18-20	17	75	8	1.5 (9)
21	23	62	15	3.4 (26)
22	25	61	14	11.3 (87)
23	27	60	14	20.1 (155)
24	34	62	5	20.9 (161)
25	46	44	9	19.4 (149)
26	58	38	4	14.0 (108)
27	63	35	2	6.0 (46)
28	73	18	9	1.4 (11)
29-34	93	0	7	0.1 (15)
Column totals	40.0 (308)	51.3 (395)	8.7 (67)	100 (770)

¹McIntyre scale indicates responses to the question, "I would like to be ...".²Percentages are calculated by row, e.g., within each SET* category.³Numbers in parentheses are the total number of votes in the respective column or row.

them. The dependent variable in each case is the *mean* response within each temperature bin.* Fits are weighted by the number of votes making up each mean response. Table 6 shows the slope, y -

*For reasons of clarity and custom, the binned mean responses are analyzed instead of all responses. An earlier paper reporting on these data showed that the regression results, in terms of slope and intercept, were identical between the two approaches, but when all responses were used, the fits were significantly poorer due to the enormous scatter inherent in any distribution of thermal comfort responses.

TABLE 5. Cross-tabulation of SET* vs. McIntyre scale in naturally ventilated buildings

SET*	% McIntyre scale votes ^{1,2,3}			Row totals
	"Cooler"	"No change"	"Warmer"	
24-26	67	33	0	2.4 (9)
27	56	44	0	2.4 (9)
28	65	35	0	5.3 (20)
29	75	25	0	7.4 (28)
30	75	25	0	10.6 (40)
31	73	25	3	19.4 (73)
32	85	14	2	17.6 (66)
33	69	24	7	15.4 (58)
34	86	12	2	15.2 (57)
35	94	6	0	4.3 (16)
Column totals	76.6 (288)	21.3 (80)	2.1 (8)	100 (376)

¹McIntyre scale indicates responses to the question, "I would like to be ...".²Percentages are calculated by row, e.g., within each SET* category.³Numbers in parentheses are the total number of votes in the respective column or row.

intercept, goodness of fit (R^2), the number of points going into the regression fit, and the neutral temperatures derived from these coefficients for the entire sample and the AC and NV building subgroups.

For all data points, the regression coefficients (slope terms) are 0.176 scale categories/°C and 0.182/°C for ET* and SET* as the independent variables, respectively. Much of the variation in the mean responses is explained by the respective models (i.e., high R^2 values of 0.91 and 0.96). Regression of the AC building data yields slope

TABLE 6. Regression of mean ASHRAE scale responses vs. temperature indices

	Independent variable	Slope (categories/°C)	Intercept	R^2	Nr. Pts.	T_n (°C)
All	ET*	0.176	-4.406	0.91	32	25.0
Air-conditioned	ET*	0.267	-6.582	0.86	21	24.7
Naturally ventilated	ET*	0.243	-6.670	0.79	20	27.4
All	SET*	0.182	-4.425	0.96	35	24.3
Air-conditioned	SET*	0.182	-4.443	0.79	28	24.4
Naturally ventilated	SET*	0.152	-3.473	0.74	22	22.8

terms of 0.267/°C and 0.182/°C for dependent variables of ET* and SET* respectively, and for the NV building data, slope terms of 0.243/°C and 0.152/°C.

The Thai AC and NV slope terms derived from ET* are roughly 50% higher than that of the whole sample, whereas the slope terms derived from SET* of the different aggregations of the data shown in Table 6 are all remarkably close.

Regression results of the disaggregated portions of the data are less robust than those of the sample taken as a whole, particularly when SET* is the independent variable. This is due in part because of the fewer number of points involved, but is not clear if other factors are also involved.

The expected temperature at which a given group would vote "neutral" (T_n) can also be estimated from regression of mean ASHRAE scale response as a function of ET* or SET*. When the regression line is plotted on a graph, the neutral temperature is the point where the regression line crosses the x -axis. It is obtained computationally by taking the ratio of the y -intercept and the regression coefficient. These results are shown in the last column of Table 6.

As a function of ET*, all respondents pooled together produce a neutral temperature of 25.0 °C, while those from the AC and NV buildings produce neutral temperatures of 24.7 °C and 27.4 °C, respectively. As a function of SET*, all the neutral temperatures are lower: 24.3 °C for the full sample, 24.4 °C for the AC buildings and 22.8 °C for the NV buildings.

Thermal acceptability

The concept of thermal acceptability has been widely debated in the literature but in practice is difficult to determine. As a generally accepted definition, ASHRAE, in the formulation of its comfort standard, *Standard 55-81*, has adopted the convention that thermal sensation votes within the central three categories of the seven-point scales (i.e., from -1 to 1) connote acceptance of the

thermal environment [6]. Along with this criterion, the ASHRAE comfort standard seeks to satisfy the thermal needs of 80% of building occupants. The standard does not seek to satisfy everybody since experiments have shown that at any thermal condition, some percentage of the building occupants will not be satisfied. Also, the standard embodies some attempt at balancing the needs of human comfort and health with the need to minimize energy costs.

ASHRAE Standard 55-81 depicts a summer thermal comfort "zone" bounded on two sides by ET* of 22.8 °C to 26.1 °C (also bounded by humidity levels for health considerations). Figure 10 is a relative frequency scatter plot of the percentage of votes at "thermal acceptability" (i.e., ASHRAE scale votes between -1 and 1) over the range of temperatures. Because the standard is defined in terms of ET*, Fig. 10 uses this temperature index in bins of 0.5 °C instead of SET*. The smooth curves are fits of the relative frequency data weighted by the number of votes in each ET* bin. By noting where a curve crosses the 80% line, one can determine the temperature bounds of the comfort zone for the sample represented in each curve. For the sample from AC buildings, the lower boundary of the comfort zone is about 22 °C and the upper boundary reaches nearly 28 °C. For the NV building sample, the lower boundary of the comfort zone is undefined, but the upper boundary is approximately 31 °C. The air velocities experienced by the respondents at each ET* bin do not vary significantly from the mean found in the respective building types (see Table 1), even at the highest ET* bins.

Discussion

Comparison with other field studies

A large number of thermal comfort field studies have been conducted, spanning decades and continents. These studies give us a context within which the Thai results can be evaluated. Here we will

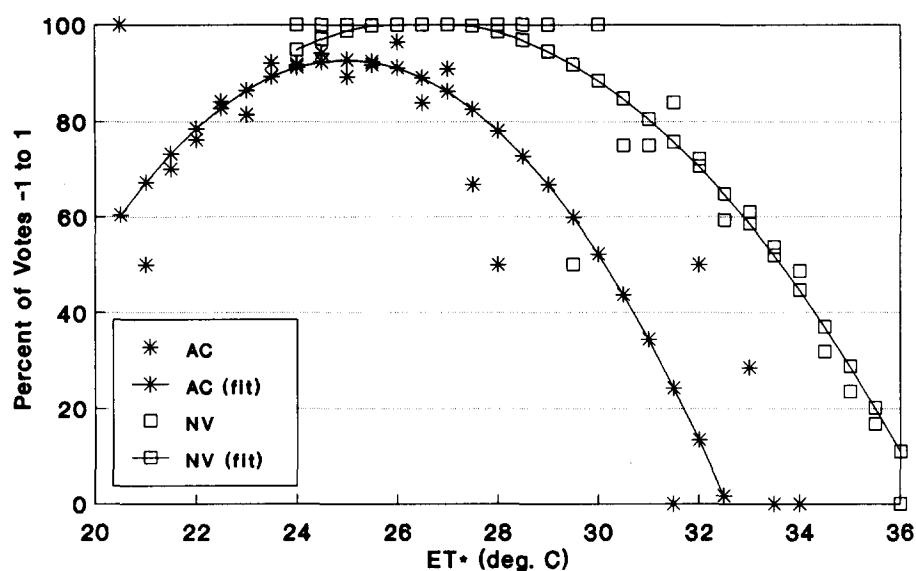


Fig. 10. Thermal acceptability in Thai offices.

focus on those field studies undertaken in tropical climates, with occasional reference to results from temperature climates.

Regression techniques are frequently employed in analyzing field study data, and results from these studies provide the primary basis for our comparisons. We caution the reader that there are some differences in how the present study formulated the regression equations as compared to earlier studies. Most of the field studies in the literature used dry-bulb or mean radiant temperature (whichever was available), for the independent variable whereas the present study employed ET^* or SET^* . We feel that our choice of independent variables more accurately captures the thermal context in a hot and humid environment than do the temperature indices most frequently used in previous studies. Undoubtedly some of the differences that appear in results are due to this distinction.

Humphreys [7] found a slope value of 0.23 scale categories/ $^{\circ}C$ from his compilation of field study data conducted in many seasons, locales, and in heated, cooled, and non-conditioned situations. The Thai slope terms are generally less than, but surprisingly close to, Humphreys slope terms given the variety of situations embodied in the latter. Schiller's [8] study of air-conditioned offices near temperate San Francisco during the summer season yielded a slope term of 0.308/ $^{\circ}C$, which in this case is based on ET^* instead of dry-bulb or mean radiant temperature. This result is in moderately good agreement with the Thai slope terms derived from the same dependent variable (i.e., ET^*) for both the AC and NV disaggregations.

Interpreting the slope term as an indicator of responsiveness or sensitivity of the sample population to changes in the temperature index, the AC and NV populations appear to be comparably sensitive to changes in either ET^* or SET^* . While Thai office workers from different thermal environments may be similarly sensitive to thermal change, the thermal levels to which they respond are different. Examination of the thermal responses on the McIntyre and ASHRAE scales as a function of SET^* (tabulated in Tables 2–5), reveals a striking similarity of voting *patterns* between the AC and NV buildings, but with a temperature "phase shift" upward of several degrees. Neutral temperatures (see Table 6), and the gaps between those of the AC and NV buildings, are good indicators of this temperature phase shift. Other tropical field studies have documented high neutral temperatures in relation to those from temperate zones. Notable among them are Ellis [9, 10] in Singapore who recorded T_n of 26.1 $^{\circ}C$ and 26.7 $^{\circ}C$ in separate studies, and Webb [11] who found 27.2 $^{\circ}C$ and Rao [12] with 26.0 $^{\circ}C$ (both also in Singapore), although substantially lower than Nicol's [13] work in Iran and India during hot seasons which yielded values of 32.5 $^{\circ}C$ and 31.1 $^{\circ}C$, respectively. These earlier tropical field study neutral temperatures were all drawn from unconditioned building environments and as such, compare favorably with the ET^* -derived neutral temperatures of 27.4 $^{\circ}C$ of the Thai NV buildings.

Likewise, Auliciems [14] found the neutral temperature of air-conditioned building occupants in tropical Northern Australia to be 24.2 $^{\circ}C$, or very close to the Thai AC T_n of 24.7 $^{\circ}C$. Other studies done in air-conditioned buildings in temperate cli-

mates generally find lower thermal neutralities, such as Schiller's average of 22.3 °C over winter and summer seasons.

It is interesting to note how large an effect normalization for non-standard clothing and air-movement had on the NV group neutral temperature – shifting it downward 4.6 °C to 22.8 °C. While this latter value is based on a slightly suspect correlation ($R^2=0.74$), it dramatizes how important clothing conventions and local air-movement are in influencing comfort in indoor climate conditions such as these.

While local wind and humidity conditions and clothing insulation do cause a marked shift in how the NV thermal environment is depicted by SET*, voting patterns do not appear to be significantly influenced by any one of these factors in isolation. When responses on the ASHRAE scale are disaggregated into two groups, above and below the mean of each factor above at a given SET* level, there was no statistical difference in the mean votes (at the 95% confidence level) of the two groups, nor did visual inspection of the voting patterns reveal any noticeable difference.

In sum, the group sensitivities to thermal change and thermal neutralities of the Thai respondents from NV buildings are generally in accordance with other field studies conducted in similar situations in the tropics. Likewise, Thai respondents from AC buildings hold similar sensitivities to thermal change and thermal neutralities to their counterparts from conditioned interior environments in tropical and temperate regions. Yet the voting patterns between the Thai AC and NV populations stand in stark contrast to one another. Part of the explanation for this lies in the discrepant humidities, air velocities, and clothing insulations exhibited between them. However, none of these factors individually emerge as a convincing determinant of this phenomenon. Something else not directly measured in our survey appears to be operating.

Do people acclimatize?

One of the more controversial issues in thermal comfort research is whether people adjust, in a psychological or physiological sense, to their thermal environment. Anecdotal evidence for acclimatization is strong. For example, this author has several times experienced a process of acclimatizing to hot and humid countries: the same thermal conditions which at first caused much discomfort, after a number of months came to feel quite comfortable. The notion of acclimatization is supported by field studies, as reviewed previously, but refuted by the climate chamber studies. Even a recent climate chamber

study conducted in Singapore by de Dear [15] found the preferred temperature of native subjects no different than that of temperate climate subjects. McIntyre [16] has criticized the chamber research as “reductionist” in failing to account for the myriad factors, some essentially unmeasurable, that are built into a state we call “comfort”.

Factors that contribute to the discrepancy between field and chamber studies have been postulated by others. Of these, most centered around inaccurate field measurements, especially the personal variables involving clothing and metabolism or non-thermal factors impinging on thermal determinations.

One factor largely ignored in thermal comfort studies is the thermal experience of the subjects outside of the chamber or office. Acclimatization, if it exists, presumably occurs through one's “background” thermal experience. It is admittedly difficult to collect such information, particularly since the time period immediately preceding the comfort study would be the most crucial.

As a rough proxy for capturing this “background” thermal experience, Thai respondents were asked to indicate the level of use of home air-conditioning, whether they never used it (coded 0), seldom (1), usually (2), or always (3). We show a breakdown of the level of home air-conditioning use by building type in Table 7. Half of the AC respondents and three-quarters of the NV respondents never used (or did not possess) air-conditioning in the home and hence, could be expected to have significant experience with exterior conditions. Of those who used home air-conditioning at all, the overwhelming majority “usually” used it. We tried to correlate the use of home air-conditioning with responses on the ASHRAE scale, but no significant correlation was found. Obviously, our question is not ideal for analyzing this issue. It is plausible, however, that the voting in the NV and AC samples differed from each other at least in part because the latter used air-conditioning in the home twice as much. The evidence from this study, in the form of significantly different voting patterns and general thermal experience, gives credence to the occurrence of ac-

TABLE 7. Use of air-conditioning in the home

Office type	Level of use of home air-conditioning			
	Never (%)	Seldom (%)	Usually (%)	Always (%)
Air-conditioned	53.1	3.5	38.3	5.1
Naturally ventilated	72.1	2.1	24.2	1.6

climatization within the respective Thai office building populations.

Neutrality vs. acceptability vs. preference

Because comfort is not directly measurable, researchers have struggled with various techniques for estimating this elusive quantity, often through indirect means or inference. Different affective terms have come to characterize the search, like ones we have already introduced, including “acceptability,” and “neutrality,” or others such as “preference,” “expectation,” or “tolerance.” To some degree, the debate pivots on subtle differences in meaning among these terms.

While the concept of “acceptability” is in wide use, mainly through the ASHRAE comfort standard, “preference” has also become a familiar concept particularly through comfort work done in climate chambers. Fanger [17] has long used the method of “direct determination” in climate chamber studies, where the subject gives frequent feedback during the course of the chamber visit in order to bring the temperature to his or her preferred level. Fanger claims, on the basis of many such measurements, that people from all thermal and cultural backgrounds prefer the same conditions within a very narrow temperature range. The McIntyre scale (see Fig. 1) was developed as an alternative means for determining thermal preference, and has been used in field studies as it was here, as well as in chamber studies.

Debate on the relationship between thermal “acceptability,” “preference,” and “neutrality” is prevalent in the literature. We can examine the interplay between the terms “acceptability” and “preference”

for Thai workers by referring back to responses on the ASHRAE and McIntyre scales shown in Figs. 8 and 9. While 51% of the respondents in AC buildings indicated “no change,” a much higher 88% voted within the central three categories on the ASHRAE Scale. Similarly, only 21% opted for “no change” on the McIntyre Scale in NV buildings, but 58% voted within the acceptability range of the ASHRAE Scale. In essence, the percentage of the sample who found the NV environment acceptable was two-thirds as large as the percentage who found the AC similarly acceptable, whereas only half as many respondents (in percentage terms) preferred the NV environment as compared to the AC environment. “Preference” is clearly a stricter standard than “acceptability.”

Figure 11 presents a means for examining the relationship between “neutrality” and “preference” in the Thai office contexts. The curves are the percentage of respondents from AC or NV buildings who preferred to be either warmer or cooler (as marked) on the McIntyre scale who also voted within a given ASHRAE scale category (shown on the horizontal axis). Where the two curves from a given building type cross indicates where the group of respondents prefers to be on the ASHRAE scale. For both building types, the respondents preferred to be closer to “slightly cool” than “neutral”. This finding has been documented elsewhere by both McIntyre [16, 18] and Rohles [19] who found that during the summer season people preferred conditions somewhat cooler than neutral (though not so much cooler as the Thai sample), but in the winter season preferred somewhat warmer conditions than neutral.

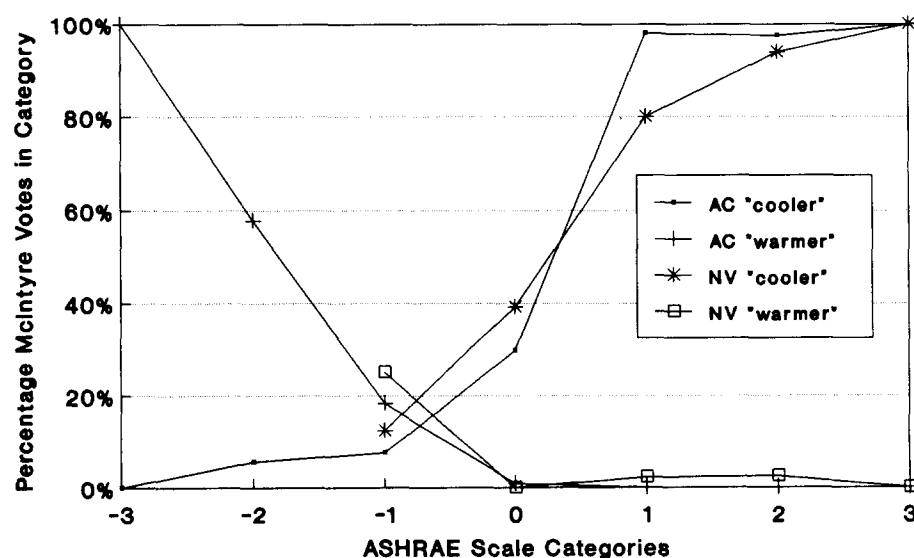


Fig. 11. McIntyre vs. ASHRAE scale votes.

Ultimately, comfort research must emerge from the quest to address ever more esoteric questions, however interesting they may be to the investigators, to address concerns of wider social import. From an energy or cost perspective, the pertinent questions are these:

- Does human performance deteriorate when the preferred or neutral thermal environment is supplanted by a merely acceptable one? If so, is the cost, in money and energy, of maintaining these more stringent thermal levels greater than the losses associated with sub-optimum human performance at the acceptable thermal level?
- Should results from field studies supersede those from chamber studies in the setting of comfort criteria for design and operation of buildings?
- Is meeting the thermal requirements of 80% of a population the ideal level for the basis of a comfort standard?

The answers to these questions have significant implications. If worker productivity was not noticeably affected by temperatures different from preferred; if field studies were used to guide thermal comfort criteria, especially in the tropics; and if the criteria were relaxed to meet the thermal needs of even a slightly smaller fraction of the building occupant population, tremendous savings potential could be harnessed [20].

Conclusions

This field study of thermal comfort in Thai offices compared two populations: one from air-conditioned offices and the other from naturally ventilated offices. From our analysis, the following conclusions can be drawn:

- Thai office workers from either interior environment are about equally sensitive to changes in thermal conditions as people surveyed elsewhere.
- The upper bound of acceptable effective temperature, defined identically to the *ASHRAE Standard 55-81* (comfort standard), is 28 °C (82.4 °F) in the AC buildings and 31 °C (87.8 °F) in NV buildings, both of which are significantly higher than the comfort standard's value of 26.1 °C (79.0 °F). Translating this finding into practice would result in abundant savings to the building sector.
- While the thermal neutralities of air-conditioned building respondents are no different from their counterparts in temperate climates, those of Thai workers from naturally ventilated offices are higher. However, lighter clothing and higher local wind explain most of the disparity.

- Like most other field studies conducted in the tropics, the present study from Thailand confirms heat tolerances higher than those found in temperate regions or in chamber studies.
- It is proposed that contrasts between the samples drawn from the two building types are evidence of acclimatization to their respective conditioned or unconditioned environments.

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References

- 1 D. A. McIntyre, *Indoor Climate*, Applied Science Publishers, Ltd., London, 1980.
- 2 *ASHRAE Handbook — 1989 Fundamentals*, American Society of Heating, Refrigerating, and Air-Conditioning Engineers, Inc., Atlanta, 1989.
- 3 *ASHRAE Handbook — 1984 Systems*, American Society of Heating, Refrigerating, and Air-Conditioning Engineers, Inc., Atlanta, 1984.
- 4 T. J. Doherty and E. Arens, Evaluation of the physiological bases of thermal comfort models, *ASHRAE Trans.*, 94 (1) (1988) 1371–1385.
- 5 A. P. Gagge, Y. Nishi and R. R. Gonzalez, Standard effective temperature — a single temperature index of temperature sensation and thermal discomfort, *Thermal Comfort and Heat Stress, Proc. CIB Commission W45 (Human Requirements) Symposium, Building Research Station, September 13–15, 1972*, HMSO, London.

- 6 ASHRAE Standard 55-1981: *Thermal Environmental Conditions for Human Occupancy*, American Society of Heating, Refrigerating, and Air-Conditioning Engineers, Inc., Atlanta, 1981.
- 7 M. A. Humphreys, Field studies of thermal comfort compared and applied, *Build. Serv. Eng.*, 44 (1976) 5-27.
- 8 G. E. Schiller *et al.*, Thermal environments and comfort in office buildings, *ASHRAE Trans.*, 94 (2) (1988).
- 9 F. P. Ellis, Thermal comfort in warm, humid atmospheres — Observations in a warship in the tropics, *J. Hygiene* (Cambridge), 50 (1952) 415-432.
- 10 F. P. Ellis, Thermal comfort in warm, humid atmospheres — Observations on groups and individuals in Singapore, *J. Hygiene* (Cambridge), 51 (1953) 386-404.
- 11 C. G. Webb, An analysis of some observations of thermal comfort in an equatorial climate, *Br. J. Ind. Med.*, 16 (1959) 297-310.
- 12 M. N. Rao, Comfort range in tropical Calcutta — A preliminary experiment, *Indian J. Med. Res.*, 40 (1) (1952) 45-52.
- 13 J. F. Nicol, An analysis of some observations of thermal comfort in Roorkee, India, and Baghdad, Iraq, *Ann. Human Biol.*, 1 (4) (1974) 411-426.
- 14 A. Auliciems and R. de Dear, Air conditioning in a tropical climate: impacts upon European residents in Darwin, Australia, *Int. J. Biometeorol.*, 30 (3) (1986) 259-282.
- 15 R. J. de Dear, K. G. Leow and A. Ameen, Thermal comfort in the equatorial climatic zone: Part I — climate chamber experiments on temperature preferences in Singapore; Part II — climate chamber experiments on temperature acceptability in Singapore, *ASHRAE Trans.*, (1991) (in press).
- 16 D. A. McIntyre, Chamber studies — reductio ad absurdum, *Energy Build.*, 5 (1982) 89-96.
- 17 P. O. Fanger, *Thermal Comfort*, Danish Technical Press, Copenhagen, 1970.
- 18 D. A. McIntyre, Three approaches to thermal comfort, *ASHRAE Trans.*, 84 (1) (1978).
- 19 F. H. Rohles, Jr., Temperature or temperament: a psychologist looks at thermal comfort, *ASHRAE Trans.*, 86 (1) (1980).
- 20 J. F. Busch, From comfort to kilowatts: an integrated assessment of electricity conservation in Thailand's commercial sector, *Ph.D. Thesis*, Energy and Resources Group, University of California, Berkeley, and *Report No. LBL-29478*, Lawrence Berkeley Laboratory, August, 1990.